

Track 25: PLC & Industrial Automation

June 2026

Track 25: PLC & Industrial Automation

MR ROKESH TECHNOLOGY — Training & Internship Programs

Skill path: Industrial Controls → Ladder Logic → PLC Programming → HMI → SCADA → Sensors → Automation Projects

Related: [Track 19 — Robotics & IoT](#) | [Track 20 — Embedded Systems](#)

Track Overview

Who this is for: Electrical, instrumentation, and automation engineers targeting manufacturing and process industries.

Prerequisites: Basic electrical knowledge; digital logic helpful.

Tools: Siemens TIA Portal / Allen-Bradley Studio 5000 (or openPLC), Factory I/O simulator, HMI panels, SCADA (WinCC/industrialSoft intro).

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	PLC awareness	Ladder logic conveyor sim
1 Month	80–100	Junior automation tech	3 PLC control projects
3 Months	250–300	Internship-ready	HMI + PLC integrated capstone
6 Months	500–600	Automation engineer entry	SCADA monitoring system
9 Months	750–900	Industry-ready	Full plant automation project

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Industrial automation overview, safety, I/O types	Safety documentation
D3–4	Ladder logic: contacts, coils, timers, counters	5 ladder programs
D5–6	PLC programming (simulator)	Motor start/stop circuit
D7–8	Sensors: proximity, photoelectric, encoders	Sensor integration lab
D9–10	Factory I/O simulation	Conveyor sorting system

1-Month Foundation

Week	Topics	Project
W1	Ladder logic, timers, counters, sequencers	Traffic light controller
W2	Analog I/O, PID intro, scaling	Temperature control loop
W3	HMI design, alarms, trends	Operator panel for process
W4	Integration capstone	Automated bottling/assembly line sim

3-Month Job Ready

Month	Topics	Projects
M1	Advanced ladder, structured text intro	Multi-station control
M2	HMI/SCADA, communication (Modbus, Profinet)	SCADA dashboard
M3	Fault handling, interlocks, capstone	Industrial automation capstone

6-Month Professional

Month	Topics	Projects
M1–M3	As 3-month track	—
M4	VFD control, motor drives	Speed-controlled conveyor
M5	Industry 4.0, IoT gateway to PLC	Remote monitoring system
M6	Capstone	Complete plant automation simulation

9-Month Advanced

Capstones: Smart factory digital twin + predictive maintenance automation system | **Level:** L5

Assessment Rubric

Component	Weight	Criteria
Weekly Quiz	20%	Ladder logic, sensors, safety
Lab Completion	40%	Programs run in simulator correctly
Project	30%	Logic correctness, HMI usability
Portfolio	10%	Project files, videos, documentation

Appendix: Mentor Notes

Emphasize lockout/tagout (LOTO) safety. Factory I/O sufficient for most tiers; real PLC hardware optional for 6-month+.